

Helium purge with O2 feedback: design

- **Date:** 2026-09-24, last updated 2026-09-25.
- **Status:** design under review. No code has been written. No PV has been created or written.
- **Measured data:** enclosure physics from the 17–24 Sep 2026 archiver data were added on 2026-09-24 (section 2.1). Values they confirm or replace are tagged MEASURED where they appear.
- **Decided 2026-09-25:**
 - implementation: synApps epid + SNL
 - IOC: 15LSS_sample_gas, port 20125
 - PV prefix: 15IDC:SampleGas:
 - the Alicat is resumed from hold automatically

1. Purpose

Replace the timer-driven helium purge (a Python script run inside a Phoebus display) with an IOC-resident controller that:

1. Purges the sample enclosure at full flow until O2 is safely below target.
2. Hands off to a PID loop that holds O2 at an operator-selected target on the least helium.
3. Keeps helium flowing through almost every fault. Zero flow is reserved for a confirmed open enclosure.

The controller runs in an IOC, so it keeps running when no Phoebus screen is open. Phoebus becomes a panel on top of it.

Evidence tags used below:

- **VERIFIED:** read from source or documentation.
- **MEASURED:** fitted to archived data from this enclosure (section 2.1 gives the source and the event count). Values from one or two events are marked "n=1" or "n=2".
- **DERIVED:** follows from reasoning.
- **PROVISIONAL:** a starting value to be tuned on the hardware.

2. Hardware and existing software

Item	Facts
MFC	Alicat BASIS, EPICS support <code>epics-modules/ip ipApp/Db/Alicat_BC.db + .proto</code> (VERIFIED). Max purge flow 20 SLPM, resolution ~0.01 SLPM. Alicat ramp rate normally 3 SLPM/s.
O2 analyzer	Updates at ~1 Hz. Noise 0.8–0.9 m% O2 per 10 s sample at hold; drift +0.13 m%/min (MEASURED). A flow change reaches the reading after 60–85 s at hold flow and 7–13 s at 20 SLPM (MEASURED, section 2.1.3).
Enclosure modes	Same volume, different leak behaviour. Mode A = normal lid (~0.25 SLPM for ~1.0 % O2); mode B = collimator lid (~0.84 SLPM for 0.99 %). Mode B's O2 ingress rises steeply as flow falls (MEASURED, section 2.1.4).
Enclosure volume	42.2 ± 0.7 L from 11 purges on the weekly archive; 40.4–40.7 L from two 10 s purges (MEASURED, section 2.1.2).
Surface limit	Flow above ~2 SLPM may vibrate the liquid surface and prevent X-ray measurement. The exact PID ceiling is to be tested.
Ambient O2	The open-lid reading is not constant : 19.0–19.6 %. It peaks right after a lift (19.48–19.56 %) and relaxes to ~19.2 % over 1.5 h; one dip to 18.95 % (MEASURED, section 2.1.8). Use 19.4 % as the model value.

Live PV names (from the user, 2026-09-24):

Signal	PV	Notes
O2 level	15IDC:D1Dmm_calc.VAL	A calc record, apparently the analyzer's output read through a DMM channel
MFC	15IDC:Alicat1:	So \$(P)\$ (R) = 15IDC:Alicat1: . Writes go to 15IDC:Alicat1:Setpoint; Setpoint_RBV is the readback.

The user owns the Alicat IOC and approves program writes to it. **Until further notice, no tool may read-modify or write these PVs:** an experiment is running. All development and testing happen against a simulator on non-colliding PV names (section 8).

Facts from Alicat_BC.db that the design depends on (VERIFIED):

- \$(P)\$ (R)Setpoint is disabled while Running_RBV = 0 (SDIS/DISV). While the controller is on hold (HLD/EXH), setpoint writes are silently dropped.
- Flow_RBV, Setpoint_RBV, Valve_RBV, Gas_RBV and Status update at 1 Hz from one poll.
- RampRate is written as SR <value> 4, and RampRate_RBV is readable.
- Run sends C, which resumes from hold.
- FlowUnits_RBV reports the units string.

Production Alicat IOC (from the user's iocIpExample folder, 2026-09-25):

- start_Alicat runs Alicat.cmd in ip-R2-22/iocs/ipExample/iocBoot/iocIpExample. It loads \$(IP)/ipApp/Db/Alicat_BC.db with P=15IDC:, R=Alicat1:, ID=A, PORT=serial1.
- The link is asyn IP to a terminal server at 10.54.170.22:8886, with \r terminators.
- Upstream ip R2-22 has no Alicat support. The deployed Alicat_BC.db and .proto in the local ip-R2-22 tree are **identical to upstream ip master** (diffed 2026-09-25).
- Production module versions: asyn R4-44-2, calc R3-7-5, StreamDevice 2-8-24, seq R2-2-9. Base is at /usr/local/epics/base, version not yet known.

Measured from alicat_trace (asyn trace, 2026-06-18, ~3.6 min; VERIFIED on the wire):

- Poll cycle: A → A +27.9 +00.70 +000084.17 +00.47 +100.00 He HLD. The fields are temperature, flow, total, setpoint, valve %, gas and status, matching the getTemperature protocol. This is followed by A SR → A +1.00000 SLPM 4 sec.
- The cycle repeats every **0.5 s**, although the deployed db's ReadSettingsScan is "1 second". Possible causes: a different db was loaded in June, or a second client was polling. Unresolved and harmless: 1 Hz is enough for this design.
- **Time-unit code 4 = seconds** (VERIFIED): the ramp reply reads "SLPM 4 sec". The ramp rate was **1.0 SLPM/s** at the time. The user reports 3 SLPM/s now.
- **Status was HLD** throughout: valve held at 100 %, flow 0.70 → 0.63 SLPM against a 0.47 setpoint. So Running_RBV = 0 and Setpoint writes would be silently disabled (SDIS). This is exactly the case the PRECHECK Run step (4.1) exists for, and it happens in practice.

Writes this design makes to **existing** PVs. They were approved by the user, who owns the Alicat IOC, on 2026-09-24:

- \$(P)\$ (R)Setpoint: purge, handoff, PID output and zero flow.
- \$(P)\$ (R)RampRate: clamped only when out of range (section 5.3).
- \$(P)\$ (R)Run: resume from hold.

Nothing else existing is modified.

2.1 Measured enclosure physics (archiver data, 17–24 Sep 2026)

Source: `analysis/` in this repo (README.md, figures/fig1–fig13, raw CSVs in `analysis/data/`). `python analysis/run_all.py` regenerates every number here from the CSVs.

Data	PV	Sampling
Weekly O2, 17–24 Sep	15IDC:D1Dmm_calc	600 s
Weekly setpoint	15IDC:Alicat1:Setpoint_RBV	on change
Weekly flow	15IDC:Alicat1:Flow_RBV	~180 s
High-res O2, 24 Sep	15IDC:D1Dmm_calc	10 s, 00:59–04:59 and 05:46–13:46
High-res flow and setpoint, 24 Sep	as above	~10 s, 09:48–13:48

The clocks of the three PVs agree to 0 ± 4 s.

2.1.1 Model

One well-mixed volume V , purged with pure helium at flow F , with O2 entering through the lid seals at rate J :

```
V · dC/dt = J(F, lid) - F · C          C in % O2, V in L, F in SLPM He, t in min
J in %·L/min;   ×10 = mL O2/min
steady state:  C_ss = J / F = J[mL/min] / (10 · F[SLPM])
flow off:      dC/dt = k · (C_amb - C),   k = 0.00142 /min (τ = 11.8 h, normal lid)
```

Replaying the whole week with only these global numbers and the delivered flow gives, with no per-episode refit:

- **normal lid:** 99 % of 428 samples within $\times/\div 1.5$, median error 8 %
- **collimator lid:** 68 % within $\times/\div 1.5$, median error 32 %. The error is limited by purge edges known only to ± 90 s.

The model's known departures are listed in 2.1.5.

2.1.2 Volume and purge (MEASURED)

- **V = 42.2 ± 0.7 L** from 11 exactly-timed purges at 3, 5.2 and 20 SLPM (2.7 % rms). The two 10 s purges give **40.7 and 40.4 L**. Use **V = 41 L** in the simulator.
- **Time constant $\tau = V/F$:** 2.1 min at 20 SLPM, 53 min at 0.8 SLPM, **169 min at 0.25 SLPM**.
- **Helium to dilute air to C:** $V \cdot \ln(19.4/C)$, whatever the flow; ≈ 125 L to reach 1 %. The flow sets only the time.
- **The existing timer purge is 20 SLPM $\times$ 6.00 min.** That is 41 purges this week, fixed at 6.00 min by high-res setpoint readings caught mid-ramp. It ends at ~ 1.1 % O2 at the sensor. **Purges used 70 % of the week's helium** (5700 of 8125 L).
- **Drop at 60 s** after the purge timer starts, for a closed box starting from air: **6.8 and 7.4 %** ($n=2$, 10 s data). An open box has not been measured.
- The Alicat reaches 20 SLPM within one 10 s sample (a mid-ramp reading of 11.85 SLPM).

2.1.3 Transport delays (MEASURED, 10 s data)

Event	Delay before O2 moves
Flow 0.25 $\rightarrow$ 0 SLPM, lid on	71 s, 60 s ($n=2$)
Flow 0 $\rightarrow$ 0.25 SLPM, lid on	O2 keeps rising for 85 s ($n=1$)

Event	Delay before O2 moves
20 SLPM purge on	13 s (edge ± 5 s); ~ 7 s in the second purge, assuming the 6.00-min program

At hold flow the loop has ~ 80 s of dead time. A flow interruption shorter than about 1 min is invisible in O2.

2.1.4 O2 ingress per lid (MEASURED, weekly data, 0.23–1.8 SLPM)

Lid (mode)	Ingress J (mL O2/min)	C _{ss}	Flow for 0.99 %	Flow for 0.5 %
normal (A)	$2.6 \cdot F^{\wedge} + 0.04$ ($\times/\div 1.27$): flat	0.26/F %	0.25 SLPM	0.51 SLPM
collimator (B)	$7.1 \cdot F^{\wedge} - 0.87$ ($\times/\div 1.31$): falls with flow	$0.71 \cdot F^{\wedge} - 1.87$ %	0.84 SLPM	1.21 SLPM

- **Mode B needs over-pressure to seal.** At 0.25 SLPM its ingress is 23–30 mL/min, and O2 climbs toward ~ 9.5 % with $\tau = 2.8$ h. That happened three times this week.
- **Seal force spread.** Within a mode, J scatters $\times/\div 1.3$ between closures, attributed to how hard the lid hardware is tightened. Two episodes fit neither family.
- **Telling the lids apart** is only possible at low flow (≤ 1 SLPM). Above ~ 2 SLPM the two families converge.
- **Flow off, normal lid.** Ingress at 1 % O2 is 10.7 mL/min, $4\times$ that with 0.25 SLPM flowing. The measured slopes are 0.024 and 0.033 %/min ($n=2$ at 10 s), against 0.026 %/min from the 7 h run on 21 Sep. The collimator lid's flow-off rate is unmeasured.

2.1.5 Mixing and the sensor zone (MEASURED, $n=2$)

- **Imperfect mixing during the purge.** The 20 SLPM decay is exponential to ± 3 %, but the apparent volume $F/(-\text{dln}C/\text{dt})$ rises from 39 to 42 L through the purge. The residual has the same shape in both purges.
- **Post-purge settling.** At the hold flow the reading keeps falling by **0.14–0.17 % with $\tau \approx 65$ s** (64 and 66 s). One well-mixed volume would take 169 min. So the sensor sits in a zone that lags the bulk by about a minute.
At purge end, the sensor reads ~ 0.15 % above where it will settle.
- **Slow tails.** Some long post-purge tails (weekly data) decay more slowly than V/F, suggesting a slow O2 reservoir (sample, plastics). This is not modelled.

2.1.6 Sealed-lid O2 behaviour: what the lid-open detector must ignore (MEASURED)

Situation	Largest rise rate	Level reached
Hold, any lid, flow on (10 s data, 12 h)	0.0008 %/s (10 s difference)	—
Flow off, lid on, final 20 s before a lift (clamps loosened?)	0.0039 %/s	1.16 %
Flow off, lid on, 7 h (21 Sep, weekly)	≤ 0.0005 %/s	9.2 %
Collimator lid starved at 0.25 SLPM (weekly)	≤ 0.002 %/s	5.7 % and rising
All sealed 600 s intervals, week	0.17 %/min = 0.0028 %/s	—

2.1.7 Lid removal (MEASURED, $n=2$ at 10 s; 32 more lifts at 600 s)

	Lift A (02:59)	Lift B (11:52)
last sealed $\rightarrow$ first big sample	1.16 $\rightarrow$ 16.48 %	1.85 (caught mid-rise) $\rightarrow$ 16.41 %
10 s difference at the lift	1.53 %/s	1.46 %/s
fast time constant	≤ 5.6 s (bound)	5.75 s (fitted; predicts the next sample to 0.06 %)
peak instantaneous slope (model)	—	~ 3.2 %/s
crosses 10 % after onset	≤ 10 s	~ 3.8 s (DERIVED from τ)

	Lift A (02:59)	Lift B (11:52)
> 18.5 % / > 19.0 % after onset	20 s / 30 s	20 s / 30 s
flow switched off before the lift	262 s	76 s

- The two lifts overlay almost exactly. A ~6 s exponential carries 97 % of the step, followed by a 1–2 min tail.
- In the weekly data, 31 of 32 lifts had the flow switched off first, and none of 32 was caught mid-rise at 600 s.
- **One lift was never seen at 600 s** (24 Sep, open 2.0 min, then purged). Only 10 s data show it.

2.1.8 Open-lid reading (MEASURED)

- It peaks right after the lift (19.48 and 19.56 %), then relaxes, not linearly, to 19.19 % over 1.5 h.
- One ~1 min dip reached **18.95 %** with the lid off (breath, or a helium pocket released).
- The weekly 600 s open-lid samples read 19.03–19.58 %, with occasional 18.7–19.0 % readings while the flow was off.
- **Any open/closed threshold near 19 % is unsafe.** The design's thresholds (10 %, 9 % arming, 18 % drop-check skip) all clear this range.

2.1.9 Setpoint readback caveat (MEASURED, archiver level)

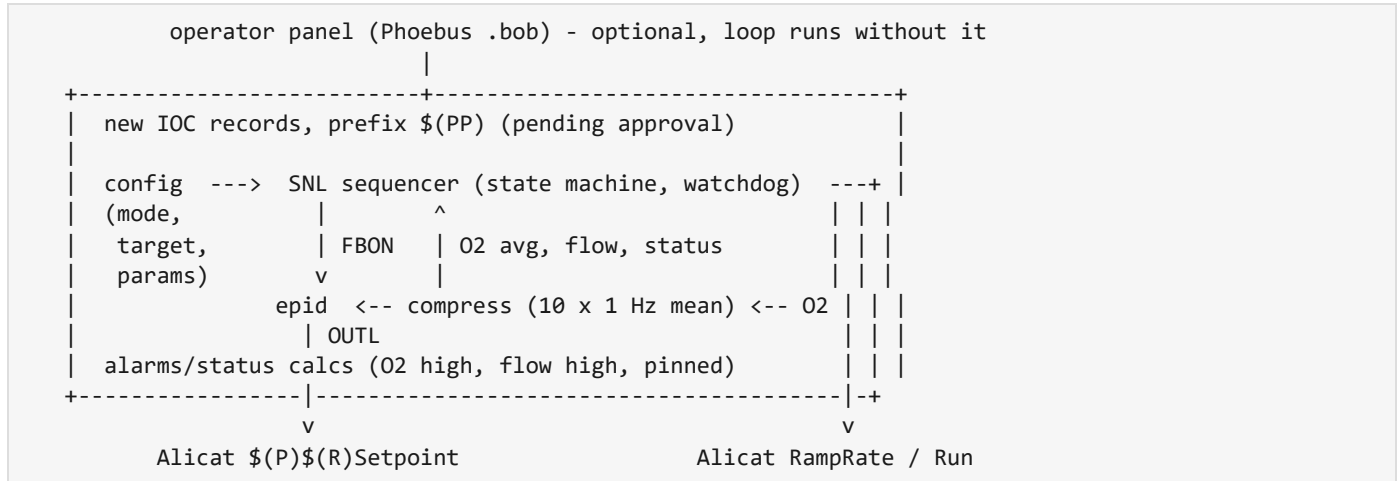
- In the archiver, Setpoint_RBV **missed 25 of 41** timer purges and the 24 Sep 11:51:02 flow-off. Flow_RBV and the O2 both show them.
- In the high-res export it read **0.43 and 19.55 SLPM mid-ramp**, so it may report a ramping, not final, setpoint.
- Whether the PV itself or only its archiving misses the events is unknown. The flow-mismatch rule (5.1) compares Setpoint_RBV with Flow_RBV, so this matters (section 9).

2.1.10 Reading noise at 1 Hz (MEASURED 2026-09-25; analysis/noise_analysis.py, fig 14)

- **Record:** 30 min at 1 Hz, normal lid, setpoint 0.25 SLPM (Flow_RBV 0.24–0.26). O2 was 0.905 → 0.932 %, and the trend was removed before analysis.
- **White sensor noise: 0.84 m%** per 1 Hz sample.
- **Trend: real enclosure O2** (user; corrected 2026-09-25, since "wander" meant this trend). +0.83 m%/min at 0.25 SLPM against a leak-model +0.41 m%/min, so an ingress 7 % above nominal explains it.
- **Slow correlated component, NOT bulk O2:** an Ornstein–Uhlenbeck process, **σ 0.67 m% (0.073 % of level), τ 79 s**.
 - It is the Allan-deviation floor of 0.32–0.44 m% from 10 to 300 s.
 - The box mixes on V/F $\approx$ 164 min, so the bulk cannot move this much in ~80 s, and it cannot step as at 16.5 min. The component is analyzer noise or local sensor-zone mixing, and the simulator applies it to the reading only.
 - Model fit: rms log error 0.04, against 1.14 for white noise alone.
- **Averaging:** the 20-sample mean floors at 0.59 m%. The 10 s archive (0.95 m% per sample over 36 windows) agrees with the model (0.87 m%).
- **Consequence for the controller:** the reading scatters by ~1 m%, 20× inside the ± 20 m% tolerance. The steady-state wiggle near target is this measurement noise, **not a loop oscillation**. Simulator comparison, scored on **bulk** O2 (revised after the correction above):
 - **Current** (10-sample mean, one gain set): bulk std 0.57 / 0.63 m% (normal / collimator lid), with 31 / 40 MFC changes per hour.
 - **Fine band 0.01 %, KP $\times$ 0.5** (Deep admin, off by default): bulk std 0.60 / 0.66 m%, with **7 / 11** changes per hour. A 30 % worse seal peaks at 12 / 19 m% bulk, still in band.
 - **Heavier smoothing** (30 samples) is worst: 0.79 / 0.83 m%.

- **Decision for the user:** the fine band costs almost nothing in bulk stability and cuts MFC moves $\sim 4\times$. The band width must be absolute and several times the ~ 1 m% scatter (≥ 0.005 %), not a percentage of target.

3. Architecture



Module	Implementation	Responsibility
Configuration	ao/mbbo records, autosaved	Mode, target, tolerance, Δ , per-mode gains, limits, thresholds
Averaging	base compress record, N-to-1 mean, N=10	Turns the 1 Hz O2 reading into the 0.1 Hz input for epid
PID	synApps std epid record	Regulation. FBON is driven only by the sequencer.
Sequencer	SNL program (seq module)	States, transitions, pre-checks, overrides, watchdog
Status/alarms	calc/calcout records with HIGH/HIHI	O2 high, flow high, PID pinned at its limit, last action, override counter
Persistence	autosave	Parameters survive IOC reboot

Keeping the watchdog in the sequencer keeps every timing rule in one readable file.

3.1 Two stations, one IOC (requested 2026-09-25)

15LSS_sample_gas hosts one independent controller **per station**. Each station has its own records, epid, state machine, parameters and autosave file.

```

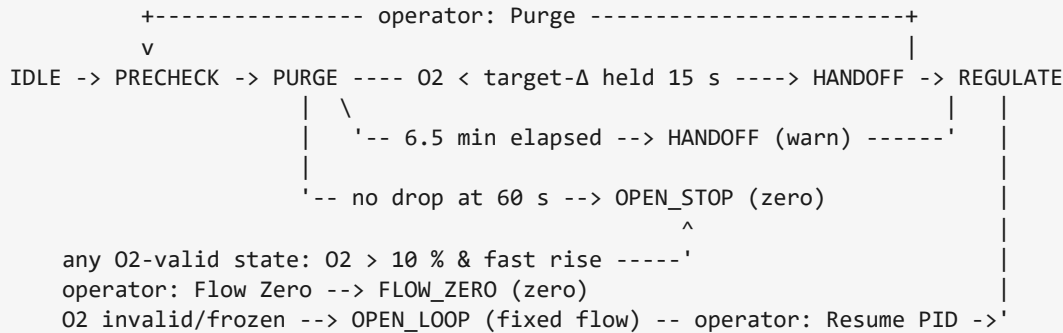
# st.cmd (sketch)
dbLoadRecords("db/sampleGas.db", "P=15IDC:SampleGas:,MFC=15IDC:Alicat1:,O2=15IDC:D1Dmm_calc,CYL=")
dbLoadRecords("db/sampleGas.db", "P=15IDE:SampleGas:,MFC=<15IDE Alicat>,O2=<15IDE O2 PV>,CYL=")
iocInit
seq sampleGas, "P=15IDC:SampleGas:,MFC=15IDC:Alicat1:,O2=15IDC:D1Dmm_calc"
seq sampleGas, "P=15IDE:SampleGas:,MFC=<15IDE Alicat>,O2=<15IDE O2 PV>"

```

- **Reentrant SNL.** The program is compiled reentrant (option `+r`), so two instances run side by side with no shared variables.
- **Adding a station** means two lines in `st.cmd`, not new code.
- **Per-station enclosure modes.** Each station has its own mode list and parameters in Deep admin. 15IDE's enclosures will have their own base flows.

- **Trade-off.** One process means an IOC crash stops both stations' controllers at once. In each case the Alicats hold their last flow, and there is one restart. Accepted in exchange for one IOC to build, deploy and maintain.
- **15IDE PV names:** still to be supplied.

4. States



State	Flow	epid FBON
IDLE	unchanged (sequencer does not write)	0
PRECHECK	unchanged	0
PURGE	20 SLPM	0
HANDOFF	expected flow for the mode, then FBON=1	0 → 1
REGULATE	PID output, within DRVL .. DRVH	1
OPEN_LOOP	expected flow for the mode, fixed	0
FLOW_ZERO	0 (operator request)	0
OPEN_STOP	0 (enclosure confirmed open)	0

4.1 PRECHECK: correct and warn, never refuse

A purge must not fail because of a bad configuration. Each check either corrects the problem or warns, then the purge proceeds.

Check	Action
Running_RBV = 0 (hold)	Write Run, wait 3 s, re-check. If still on hold: MAJOR alarm, but stay in PURGE so that flow starts the moment hold clears.
RampRate_RBV > 5 or = 0	Write 5 to RampRate. MINOR alarm, log.
Gas_RBV ≠ He	MINOR alarm only. Do not change the gas table.
FlowUnits_RBV not SLPM	MAJOR alarm. The ramp and flow numbers would be in the wrong units. The purge still proceeds.
O2 invalid or frozen	Proceed in blind purge mode (section 4.3).

Treating 0 as "fastest" assumes the Alicat's 0 means ramping is off. **Confirm this in the BASIS manual.**

4.2 PURGE

1. Record the O2 reading at the start, O2_start.

2. Write 20 SLPM.

3. Start the purge timer when `Flow_RBV` first reaches 95 % of 20 SLPM. At 3 SLPM/s that takes about 6.5 s.

4. **Lid check, onset-based and faster** (REVISED again 2026-09-25; supersedes the fixed-window version directly below):

- **Wait for the decay onset:** the first reading $\geq$ `lidOnsetFrac` (2 %, relative) below the purge-start O₂. With the lid on this comes ~10–13 s after full flow. The measurement window starts wherever the decay actually starts, so uncertainty in the 7–18 s transport delay does not matter.
- **No onset within `lidOnsetMax` (30 s)** → no decay → **OPEN_STOP**.
- **Otherwise, measure over `lidWindow` (15 s) after onset:**
 - rate ratio = $k_{\text{obs}} / (F/V)$
 - curvature = (decay rate in the 2nd half) / (1st half)
- **Why curvature:** with the lid on, the decay is exponential, so the curvature is $\approx$ 1.0–1.1. An open box also dilutes quickly at first but then levels off.
- **Decide:** open if ratio < `openSlopeFrac` (0.5) **or** curvature < `lidCurvMin` (0.8).
- **Simulator:** decides **~26 s after full flow** with the lid on (was ~60 s), with 0 false trips in 20 closed-lid purges.
 - **Closed lid:** ratio 0.95, curvature 1.10.
 - **Open box**, by air-exchange multiplier:
 - $\times 1$: 0.11 / 0.27
 - $\times 0.5$: 0.33 / 0.51
 - $\times 0.3$: 0.52 / 0.69, caught by curvature
 - $\times 0.2$: 0.63 / 0.78, caught by curvature
 - $\times 0.1$ passes: that box is effectively sealed.
 - The earlier 10 s window without curvature missed $\times 0.3$.

*Previous version: **Lid check from purge kinetics** (REVISED 2026-09-25 at the user's direction; superseded the `MinDrop` rule below):*

- With the lid on, O₂ decays as $\exp(-F \cdot t/V)$. With the lid open, the helium rises out of the enclosure and the decay is **at least 2× slower** (user's physical expectation).
- **Measure** the decay rate $k_{\text{obs}} = \ln(C(20 \text{ s}) / C(60 \text{ s})) / 40 \text{ s}$, with times counted from full flow. The window starts after the ~10 s transport delay and sensor lag at 20 SLPM.
- **Compare** it with the lid-on rate $k_{\text{exp}} = F/V$ (20 SLPM / 41 L = 0.0081 /s).
- **Decide:** if $k_{\text{obs}} < \text{openSlopeFrac}$ (0.5) $\times k_{\text{exp}} \rightarrow$ **OPEN_STOP**.
- The measure is independent of the starting level, so it does not depend on ambient O₂.
- The skip below `dropSkipLevel` (18 %) is kept as a second guard.
- **Deep-admin parameters:** `V` (41 L), `kinT1` (20 s), `dropCheckTime` (60 s), `openSlopeFrac` (0.5).
- **Simulator result:** closed lid 100 % of k_{exp} (reality: apparent `V` 39–42 L, so ~95–105 %). Open lid 0 % with the default model. The open box would only pass if its air exchange were ~10× weaker than modelled (`openMix` $\leq$ 0.05). The open-lid purge test settles this.

*Superseded rule, kept for reference: **Drop check at 60 s** (on the purge timer):*

- Skip it if `o2_start` < 18 %. O₂ that low proves the enclosure was closed.
- Otherwise require `o2_start - o2_now` $\geq$ `MinDrop`. Default `MinDrop` = 1.0 % O₂ (PROVISIONAL; tune from logged purges).
- If the check fails: **OPEN_STOP**.
- Measured basis: a closed box starting from air has dropped **6.8–7.4 %** by this point ($n=2$; 2.1.2), so 1.0 % leaves ~7× margin. The response of an **open** box to 20 SLPM is unmeasured. The lid-open test (section

- 8) must confirm it stays below 1 %.
- The 18 % skip threshold clears every open-lid reading seen ($\geq 18.67\%$) by at least 0.67 % (2.1.8).

5. **Early handoff** (REVISED 2026-09-25):

- **Rule:** go to HANDOFF when the **lag-corrected** O2 is below target $-\Delta$ for `handoffHold` (5 s).
- **Correction:** $\text{estimate} = \text{reading} \times \exp(-k \cdot \text{purgeLag})$, where k is the decay rate measured by the lid check (fallback F/V) and `purgeLag` = 12 s (transport plus sensor lag at 20 SLPM).
- **Why:** at 20 SLPM the analyzer reads $\sim 10\%$ high. The old uncorrected rule ($\Delta 0.15\%$, 15 s hold) handed off with the true O2 at $\sim 0.78\%$ while the reading said 0.87 %. O2 then dipped to 0.76 %, and took 19 min (collimator) to 72 min (normal lid) to return to $\pm 0.02\%$.
- **Result:** with $\Delta = 0.01\%$ (user, 2026-09-25) plus the correction, the dip is $\sim 0.91\%$, back in band after 12 min (collimator) and 30 min (normal lid).
- **Default now $\Delta = -0.04\%$** (user, 2026-09-25): hand off 0.04 % above target, on top of the lag correction. After a purge to 0.99 %, O2 stays within 0.960–1.036 % (normal lid) and 0.954–1.037 % (collimator lid), back in band after 15 and 9 min.
- **Negative Δ** (hand off *above* target) is allowed, as the user suggested. $\Delta = -0.10\%$ with no correction performs about the same at a 1 % target. The correction scales better at other targets because it is proportional.
- Δ previously defaulted to 0.15 %.

6. **Timeout** (REVISED 2026-09-25: scales with the target):

- **Formula:** $\text{timeout} = \text{purgeTimeoutMargin} (1.3) \times [(V/F) \cdot \ln(C_{\text{start}} / (\text{target} - \Delta)) + \text{purgeLag} + \text{handoffHold}]$, clamped to 120–1800 s.
- **After the lid check** it is recomputed from the measured decay rate and the O2 at the check. A blind purge assumes $C_{\text{start}} = \text{ambientRef}$ (19.4 %).
- **Examples from air:** 11.5 min for a 0.3 % target, 8.2 min for 0.99 %, 5.2 min for 3 %. In the simulator every purge from 0.3 % to 3 % now hands off on O2, not on the timeout.
- **On timeout:** go to HANDOFF anyway and raise a MINOR "purge incomplete" alarm.
- The fixed 6.5 min timeout is superseded.

Measured timing conflict (DERIVED from 2.1.2, needs a decision; section 9).

- From air, reaching target $-\Delta = 0.84\%$ at 20 SLPM takes **6.4–6.6 min of flow** ($V = 40.5\text{--}42.2\text{ L}$). Add $\sim 13\text{ s}$ of sensor delay and the 15 s hold.
- With the defaults, the 6.5 min timeout will therefore often fire first, and "purge incomplete" becomes a routine alarm.
- At 6.5 min the sensor reads 0.78–0.89 %. It then settles a further $\sim 0.15\%$ within $\sim 3\text{ min}$ (2.1.5), so the bulk is already lower than the reading.
- Options:
 - raise the timeout to $\sim 7.5\text{ min}$;
 - hand off on target $-\Delta$ with a smaller Δ ;
 - accept the alarm.
- Each extra minute at 20 SLPM costs 20 L of helium.

4.3 Blind purge (O2 unavailable)

Run 20 SLPM for the full 6.5 min, with no drop check and no early handoff. Then go to OPEN_LOOP. This reproduces today's timer behaviour.

4.4 HANDOFF

1. If `RampRate_RBV` < 2 and $\neq 0$, write 2. MINOR alarm, log.
2. Write the mode's expected flow to `Setpoint`.
3. Wait until `Flow_RBV` is within tolerance of it.
4. Set `FBON` = 1.

`epid` initialises its integral from `OUTL` when `FBON` goes 0→1, so the loop starts at the expected flow instead of 20 SLPM (VERIFIED, `devEpidSoft.c`).

4.5 REGULATE

`epid` runs every 10 s on the 10-sample O₂ mean. The sequencer watches alarms and the lid-open condition. There is **no automatic re-purge**: a re-purge happens only when an operator presses Purge. Headroom for upsets comes from the tested PID ceiling (section 6).

4.6 Lid-open detection

- **Trigger**: (O₂ > 10 % for 5 consecutive seconds) **AND** (rise rate $\geq$ `LidSlope` at any point in the trailing 60 s). It applies in any state where O₂ is valid and the sequencer is controlling flow.
 - The 5 s filter rejects single-sample glitches.
 - An opened lid raises O₂ faster than any other process, so the rate term rejects slow drifts that happen to cross 10 %.
 - Rise rate = (O₂ now – O₂ 10 s ago) / 10 s, computed at 1 Hz.
 - The rate is taken as a **trailing-60 s maximum** because O₂ plateaus near ambient after the lid opens. The instantaneous rate may already be ~0 by the time the 5 s threshold filter completes.
 - `LidSlope` default: 0.2 % O₂/s (PROVISIONAL; set from the deliberate lid-open test, section 8).
- **Consequence of the AND**: a lid only cracked open, leaking slowly, will not trip this detector. Instead the PID rises to `DRVH`, and the "flow too high / PID pinned" MAJOR alarm fires. Helium use in that case is bounded by `DRVH` (≤ 2 SLPM), not 20 SLPM.
- **During PURGE**, the detector arms only after O₂ has first fallen below 9 %. Otherwise a purge starting from air would trip it immediately.
- **Action**: `OPEN_STOP`.

This covers the case where the operator forgets Flow Zero before opening the enclosure.

4.6.1 Measured basis for inferring the lid state (MEASURED, 2.1.6–2.1.8)

The two terms separate a lift from every sealed behaviour recorded:

Quantity	Lid lifted (n=2 at 10 s)	Sealed, worst case seen	Margin at the design threshold
10 s rise rate	1.46–1.53 %/s	0.0039 %/s (flow off, clamps being loosened)	<code>LidSlope</code> 0.2 %/s: 7× below lifts, 51× above sealed
O ₂ level	> 18.8 % within 20 s, > 19 % within 30 s	9.2 % (7 h flow off); 5.7 % and rising (collimator lid starved)	10 %: lifts cross in ~4 s; nothing sealed reached it
Both together	yes, ~5–10 s after onset	never	—

- **Neither term alone is enough**. A sealed box with the flow off does creep past 10 %: 9.2 % after 7 h on 21 Sep, heading for ambient. But it does so at ≤ 0.0005 %/s. A cracked lid or a starved collimator lid is similarly slow. The rate term rejects all of these.

- **Detection latency** (DERIVED from the measured $\tau = 5.75$ s): O2 crosses 10 % ~4 s after the lid starts to lift, and the 5 s filter completes ~9 s after. At that point the trailing-60 s maximum of the 10 s difference is ~1.5 %/s, 7× over `LidSlope`. **Expect OPEN_STOP within ~10 s of a lift.** The 10 s archive shows the reading at > 16 % one sample after onset in both lifts.
- **LidSlope = 0.2 %/s is supported by the data and may stay.** Anything from ~0.05 to ~0.5 %/s keeps at least 3× margin on both sides. At the analyzer's 1 Hz, the 10 s difference behaves the same way. The peak instantaneous slope is ~3.2 %/s.
- **Pre-lift warning (optional, n=1).** In the 20 s before lift A, O2 rose at 0.16 %/min, 5× the flow-off leak-in. It could serve as an early "clamps released" hint, but one event is not enough to act on.
- **Not covered:** the collimator lid has no 10 s record. The lift kinetics are expected to be similar (same box and sensor), but this is unverified.

4.6.2 Continuous hold monitor (DECIDED 2026-09-25: resume automatically)

- **Why.** In `HLD` the Alicat holds its **valve** at a fixed opening and ignores the setpoint. The 2026-06-18 trace shows the valve at 100 % and 0.7 SLPM flowing against a 0.47 setpoint. Meanwhile `Setpoint` writes are skipped (`SDIS` on `Running_RBV`).
- **Rule.** Users stop helium only by setting flow to 0, never by holding the valve.
- **Where it applies.** In every state where the sequencer owns the flow: `PURGE`, `HANDOFF`, `REGULATE`, `OPEN_LOOP`, `FLOW_ZERO` and `OPEN_STOP`. The zero-flow states are included, because a held valve keeps helium flowing whatever the setpoint says.
- **Trigger.** `Running_RBV` = 0 for 3 s.
- **Action:**
 1. Write `Run`.
 2. Once `Running_RBV` = 1, **re-send the last commanded setpoint** by writing it again with processing. While the record was disabled, its `VAL` may have changed without ever being sent to the device. Whether `epid` rewrites an unchanged output every scan is **unverified**. The explicit resend makes recovery certain either way.
 3. MINOR alarm "MFC was on hold, resumed". Log it and increment `OverrideCount`.
- **If it's still on hold after 3 attempts 10 s apart:** MAJOR "MFC on hold, cannot resume". Keep retrying every 60 s.
- **IDLE** is excluded: there the controller does not own the flow.
- The deployed `Alicat_BC.db` and `.proto` are identical to upstream ip master (diffed 2026-09-25).

4.7 IOC restart

The Alicat holds its last setpoint on its own, so helium keeps flowing if the IOC dies (DERIVED from the device being a standalone controller). On restart:

Condition	Result
O2 valid, < 10 %, and <code>Setpoint_RBV</code> > 0	<code>REGULATE</code> . <code>epid</code> picks up from the current setpoint without a bump.
Setpoint is 0	<code>IDLE</code>
O2 invalid	<code>OPEN_LOOP</code> if setpoint > 0, otherwise <code>IDLE</code>

5. End-state table (living; add to it as the design evolves)

Situation	Flow	Alarm	Recovery
No O2 drop at 60 s (confirmed open enclosure). A closed box drops 6.8–7.4 % by then (MEASURED)	0	MAJOR "no O2 drop, enclosure open?"	Operator closes enclosure, presses Purge
O2 > 10 % for 5 s AND fast rise (lid opened without Flow Zero). Fires ~10 s after a lift (DERIVED from measured kinetics, 4.6.1)	0	MAJOR "enclosure opened, flow stopped"	Operator closes enclosure, presses Purge
Operator Flow Zero. With the lid kept on, O2 starts rising 60–85 s later at 0.024–0.033 %/min, about +0.5 % in 20 min (MEASURED, normal lid)	0	none	Operator presses Purge
Lid lifted and replaced between purges faster than the archive samples (seen once: 2 min open, 24 Sep 11:52)	follows the operator's Flow Zero / Purge	none	Operator presses Purge. Archive O2 at 1 Hz or 10 s: a 600 s archive can miss the whole event
Target – Δ not reached at 6.5 min	PID (via HANDOFF)	MINOR "purge incomplete". The flow-high alarm will likely follow.	Automatic
O2 analyzer invalid or frozen	Expected flow, fixed (OPEN_LOOP)	MAJOR "O2 lost, running blind"	Operator presses Resume PID once O2 returns
MFC went on hold (any controller-owned state, 4.6.2)	valve held at its last opening, until resumed	MINOR "MFC was on hold, resumed"	Automatic: Run, then re-send the setpoint
MFC on hold and 3 Run attempts failed	valve held (may be flowing, may be closed)	MAJOR "MFC on hold, cannot resume"	Operator. Controller keeps retrying every 60 s.
Ramp rate out of range	unaffected (clamped)	MINOR, logged	Automatic
Gas table ≠ He	unaffected	MINOR	Operator, optional
Flow ≥ 1.5× expected	PID	MINOR "flow high, check enclosure"	Automatic
PID demand (OVAL) ≤ flowLowX (0.5) × expected flow for alarmDelay, after the settle time (typically collimator mode chosen with the normal lid fitted). Judged on the demand, not the measured flow, so an empty cylinder does not trigger it.	PID (regulating normally at the lower flow)	MINOR "flow ≤ 0.5× expected: wrong enclosure mode selected?"	Operator selects the correct mode
PID pinned at DRVH (0.01 SLPM) for 10 min with O2 below target – tolerance	PID (at minimum)	None. A log entry only: O2 too low is not an operator alarm (user, 2026-09-25). The 0.01 SLPM floor exists for a planned tighter enclosure; the current one needs far more flow.	none
Flow ≥ 2× expected, or PID pinned at DRVH for 10 min	PID	MAJOR "flow too high, check enclosure seal"	Operator

Situation	Flow	Alarm	Recovery
Controller IOC down (crash; procServ runs --noautorestart)	Alicat holds its last setpoint	None from this IOC. A heartbeat PV going stale can be watched by the alarm server or the easy-bluesky watchdog.	Operator restarts it via <code>start_ioc</code> <code>15LSS_sample_gas</code> . On restart, the rules in 4.7 apply.
Cylinder run-out forecast (median over windows, 5.5) below 24 h / 6 h	unaffected	MINOR / MAJOR "helium cylinder empty in < 24 h / 6 h (forecast)"	Operator fits a new cylinder, presses "New He cylinder fitted"
Flow not matching setpoint (see 5.1)	PID continues	MAJOR "flow mismatch: cylinder empty or MFC fault?"	Operator

Zero flow happens only for a confirmed open enclosure or an explicit operator request.

5.1 Flow-mismatch rule

DECIDED 2026-09-25: compare `Flow_RBV` with the Alicat's own `Setpoint_RBV`. The user confirms the live PV is reliable; the gaps seen in 2.1.9 are an archiving artefact. Writes that the Alicat drops while on hold are caught by the hold monitor (4.6.2), not by this rule. The allowed ramp time is measured from the moment `Setpoint_RBV` changes.

When the setpoint changes, compute:

$$\text{allowed_s} = |\text{new_setpoint} - \text{Flow_RBV}| / \text{RampRate_RBV} + 5 \text{ s}$$

(If `RampRate_RBV` = 0, use 5 s.)

Alarm if $|\text{Setpoint_RBV} - \text{Flow_RBV}| > \max(0.05 \text{ SLPM}, 5 \% \text{ of setpoint})$ is still true after `allowed_s`. The tolerance is PROVISIONAL.

5.1.1 Process-alarm gating: settling logic (ADDED 2026-09-25; replaces a fixed settle time)

The process alarms are "flow $\geq 1.5 \times / 2 \times$ expected", "PID pinned at max", "O2 above range / abnormally high" and "flow $\leq 0.5 \times$ expected (wrong mode?)". Each can be legitimately true while the loop is still moving to a new target, so each is gated by the state history. The mismatch, hold, lid and purge alarms have their own timing and are not gated.

- **Settling flag.**
 - **Set** on REGULATE entry and on every target change. It records the direction: O2 falling or rising toward the target.
 - **Cleared** when O2 reaches the target itself, crossing it in the settling direction ("the flag switches polarity").
 - After that, any excursion is a disturbance, and the alarms apply normally.
- **Progress while settling.**
 - **Slope:** O2 change over `stallWindow` (120 s), using the mean of `slopeAvgN` (20) samples at each end.
 - **Judging** starts after `stallGrace` (300 s: dead time plus the slope window).
 - **Stalled:** less than `progressMin` (0.0005 %/min) toward the target for `stallTime` (120 s). A stall **latches** until the target is reached or changed. Once stalled, the alarms apply at once.
- **Settle timeout.** Still settling after `settleTimeout` (2 h) → MINOR "target not reached", and the alarms apply. This catches a target the loop cannot reach, or reaches only asymptotically.
- **Steadiness for flow alarms.**
 - Flow-high and flow-low are judged only when the process is steady: PID output within $\pm \text{flowSteadyBand}$ (0.02 SLPM) over the slope window **and** $|\text{O2 slope}| < \text{o2SteadyRate}$ (0.002 %/min).

- A genuine seal or mode problem is steady. The PID winding flow down after a downward target change, or up after a purge, is not.
- **Persistence.** Flow and O₂-range alarms must hold for `flowAlarmDelay` (300 s). Other alarms use `alarmDelay` (10 s).
- **Simulator results.**
 - **No false alarms** for target steps 0.99 ↔ 0.5 / 1.5 / 2.0 % on both lids, or for purges from air.
 - **Real problems still alarm:**
 - unreachable target (normal lid, seal ×1.3, 0.3 %): timeout, then PID pinned
 - cracked lid: O₂ abnormally high at 15 min, PID pinned at 17 min
 - collimator lid in mode A: flow ≥ 2× expected after settling
 - normal lid in collimator mode: wrong mode?

5.5 Helium cylinder run-out forecast (ADDED 2026-09-25; no pressure PV needed)

- **Source:** the Alicat's own totalizer, `$(P)$ (R)Total_RBV` (standard litres, kept in the device across IOC restarts). This is **read only**; the controller never resets it.
- **New cylinder:** the operator presses "**New He cylinder fitted**" on the main panel. The IOC stores the current `Total_RBV` as the baseline (autosaved).
- **Remaining:** `cylCapacityL` (usable litres, default 8000 L, Deep admin) – litres used since the baseline. If the totalizer goes backwards, for example because someone reset it, the baseline is shifted so the count continues.
- **Forecast:**
 - Litres used are logged every 60 s, and 4 days are kept (an autosaved waveform in the IOC).
 - For each window of **3, 2, 1, 0.5 and 0.25 days**, a least-squares usage rate (L/day) gives run-out = remaining ÷ rate. Windows with less than 90 % data coverage are skipped.
 - The **spread across windows is the confidence range** shown on the main panel, e.g. "empty in 5.5–5.6 d (median 5.5 d, 5 windows)".
- **Alarms** on the median: MINOR below `cylWarnH` (24 h), MAJOR below `cylAlarmH` (6 h).
- **Simulator check:** 4 days of use (regulation at 0.99 % plus a lid cycle and purge every 6 h, ~840 L/day). At day 4 all windows predicted 5.50–5.53 d. The cylinder actually ran out 5.52 d later. Early forecasts are honestly wide: at 6 h, only the 0.25 d window exists and it had seen just one purge.
- **Pressure PV (macro CYL), once it exists:** shown alongside, and usable to cross-check `cylCapacityL`. Nothing depends on it.
- **IOC implementation:** the 1-min usage log and the per-window regressions are simple enough for the SNL program (a 5760-point array), with results published to records.

5.6 Helium ledger: cylinders and helium per user run (ADDED 2026-09-25)

- **Litre count:** monotonic, the sum of positive steps in `Total_RBV`, so a totalizer reset cannot corrupt it.
- **Events**, each stored with a timestamp and the litre count:
 - purge start (PURGE entry)
 - flow set to zero (FLOW_ZERO or OPEN_STOP entry)
 - new cylinder ("New He cylinder fitted")
- **Snapshots:** hourly snapshots of the litre count.
- **Storage:** events and snapshots are kept for `reportDays` + 10 days, persisted across IOC restarts (autosaved waveforms).
- **User run:** a series of purges. Flow-zero events between purges are sample changes within the run.
 - The run **ends at the first flow-zero after its last purge**. If there is none, it ends at the last purge.
 - It counts as finished once no purge follows for `runGap` = **60 h** (REVISED 2026-09-25).
 - **Why 60 h:** the beam has 24 h and 48 h downtimes after which the same user resumes, and users may take up to ~10 h to start a purge after the beam returns.

- **Measured from the flow-zero** that ended the segment (beam down), not from the last purge. Otherwise the hours between the last purge and the beam loss would push a 48 h downtime over the limit.
 - **"Mark start of new user run"** (Admin) splits runs at a user changeover, whatever the gap. It is needed when a new user starts within 60 h of the previous one.
 - Per run: start, end, number of purges, litres (exact, from the boundary events) and cylinder changes.
- **Report window:** `reportDays` (60) **ending at the end of the latest user run**, so idle time afterwards does not dilute it. It shows:
 - helium dispensed (L)
 - litres during user runs, clipped to the window
 - cylinders fitted
 - cylinder-equivalents (litres ÷ `cylCapacityL`)
 - the run table
- **Where it appears:** Admin screen ("Helium usage report").
- **Simulator check (scenario 19):**
 - **User A:** days 0–3, a 48 h downtime, resuming 10 h after beam return, until day 8. This stays **one run**: 23 purges, 4777 L.
 - **User B:** after a marked changeover, days 8.5–11 with a cylinder change: 2103 L.
 - **User C:** after 3.5 idle days, split automatically: 841 L.
 - **Totals:** dispensed = sum of runs = Alicat totalizer = 7721 L.

5.2 Expected flow

Model (MEASURED, 2.1.4): $C_{ss} = J(F)/F$, with ingress $J = a \cdot F^b$ per lid. So:

$$\text{expected_flow} = \text{mode_base_flow} \times (0.99 / \text{target})^n, \quad n = 1 / (1 - b)$$

Mode	mode_base_flow (for 0.99 %)	b	n	Flow for 0.5 %
A (normal lid)	0.25 SLPM	+0.04	1.0 (the original linear rule)	0.51 SLPM
B (collimator lid)	0.84 SLPM	−0.87	0.53	1.21 SLPM (the linear rule would give 1.66)

- The original rule ($O_2 \approx 20.9 \cdot L/F$, i.e. $n = 1$) holds for mode A. It **over-predicts mode B's flow change** by ~40 % at 0.5 %, because more over-pressure seals the collimator lid better.
- The sensor's ambient is 19.4 %, not 20.9 %. This only matters for absolute leak-rate numbers.
- Seal-force spread is $\times/\div 1.3$ on J , so expect `expected_flow` to be off by up to ~30 % for a given closure. The PID absorbs that.
- **Consequence for the flow alarms (section 5).**
 - The MINOR "flow $\geq 1.5 \times$ expected" threshold sits only ~15 % above a badly sealed but normal closure in mode A. Expect occasional nuisance MINORs.
 - The case that matters, the **collimator lid with mode A selected**, needs $\sim 0.84/0.25 = 3.4 \times$ the mode-A flow. It therefore lands in the MAJOR "flow $\geq 2 \times$ expected / PID pinned" row, which is the right outcome.
 - Two closures this week had $1.4\text{--}2.7 \times$ the normal-lid ingress without being collimator lids. They would sit between the two alarm thresholds.

Uses of the expected flow: the HANDOFF flow, the OPEN_LOOP flow and the flow-alarm thresholds.

5.3 Ramp-rate overrides

When	Condition	Write
PRECHECK	<code>RampRate_RBV</code> > 5 or = 0	5

When	Condition	Write
HANDOFF	$0 < \text{RampRate_RBV} < 2$	2

Both overrides are logged to `$(PP)LastAction` and increment `$(PP)OverrideCount`.

6. PID configuration (from the tuning analysis)

- **Error sign.** `epid error` = `VAL - CVAL` (VERIFIED). Too much O₂ gives a negative error, which must raise flow, so `KP < 0`. `KI > 0`. `KD = 0`.
- **Units.** `KI` is in repeats/s, i.e. $1/\tau_I$ (VERIFIED).
- **Scan and deadband.** `SCAN` = 10 s. `ODEL` = 0.01 SLPM suppresses writes below the MFC's resolution. `epid` keeps accumulating the integral internally (VERIFIED). There is no error deadband: the 0.01 % sensor noise $\times$ `KP` is already below the MFC's resolution (DERIVED).
- **Limits (REVISED 2026-09-25).**
 - **DRVL is a per-lid minimum flow, with no global floor:** normal lid **0.05 SLPM**, collimator lid **0.3 SLPM**. It is set per mode in Admin, and a new, tighter enclosure gets its own mode with its own minimum.
 - **Tested in the simulator:**
 - After a purge to 0.99 %, O₂ peaks at 1.007 % (normal lid) and 1.034 % (collimator lid). A 0.01 SLPM floor let the collimator lid overshoot to 1.12 %, because it leaks fast at very low flow.
 - Purges to 0.5–2 % settle within ± 0.02 % in 12–20 min on both lids.
 - **DRVH** = mode ceiling, **hard maximum 2.0 SLPM**, to be raised or lowered by the surface-vibration test.
 - **History:** the original `DRVL` = $\max(0.1, 0.5 \times \text{expected})$ held a normal lid in collimator mode at 0.42 SLPM.
- **Feedback delay after handoff: 0 s** (user, 2026-09-25; was 120 s). The lag-corrected handoff already accounts for the sensor settling. In the simulator a delay only added 1–4 min to settling.
- **Gain scheduling (ADDED 2026-09-25).**
 - Near the operating point the O₂ slope per SLPM is $\approx C/V$, proportional to the target. A `KP` tuned at 1 % is therefore $3\times$ too strong at 3 %: the simulator's normal lid oscillated between 0.01 and 0.4 SLPM there.
 - **Rule:** the per-mode `KP` is defined "at 0.99 %", and the IOC applies $KP \times 0.99/\text{target}$. `KI`, in repeats/s, is unchanged.
 - **Result:** steady regulation at 0.3–3 % for both lids, and $2\text{--}3\times$ faster settling at low targets. `gainSchedule = 0` turns it off.
 - In the IOC this is one calc record feeding `epid.KP`.
- **Starting gains (PROVISIONAL).** `KP` ≈ -0.4 SLPM per %O₂, `KI` ≈ 0.0033 /s. Replace with per-mode values from bump tests, using SIMC tuning with $\tau_c = 2\theta$.

6.0 Process model at the hold flow (MEASURED inputs, DERIVED gains)

	Mode A (normal lid, 0.25 SLPM)	Mode B (collimator lid, 0.8 SLPM)
Steady gain $K = dC_{ss}/dF$	–3.8 %O ₂ per SLPM	–2.5 %O ₂ per SLPM
Time constant $\tau = V/F$	169 min	53 min
Dead time θ (transport ~80 s + half the 10-sample mean + scan)	~95 s	~95 s
τ/θ	~107 (effectively integrating)	~33 (effectively integrating)
Integrating slope $k' = K /\tau$	$3.8e-4$ %/s per SLPM	$8.0e-4$ %/s per SLPM
SIMC (integrating), $\tau_c = 2\theta$: $K_c = 1/(k'(\tau_c + \theta))$	9.3 SLPM/%	4.4 SLPM/%
$\tau_I = 4(\tau_c + \theta)$	1140 s	1140 s

	Mode A (normal lid, 0.25 SLPM)	Mode B (collimator lid, 0.8 SLPM)
→ KP, KI	-9.3, 8.8e-4 /s	-4.4, 8.8e-4 /s

- **Retuned 2026-09-25 in the simulator.**
 - **Mode B now uses** $K_P = -10$, $K_I = 1.4e-3$. With the old $-4.4 / 8.8e-4$ it overshoot to $\sim 1.06\%$ after disturbances and stayed outside $\pm 0.02\%$ for 16–23 min.
 - The table above used $\theta \approx 95$ s. The full model's effective dead time is ~ 60 s for mode B (transport 30 s + sensor-zone lag 19 s + averaging and scan 10 s) and ~ 155 s for mode A.
 - With the new gains, a 30 % worse lid seal stays inside $\pm 0.02\%$, and a target step settles in ~ 5 min. They were checked at targets 0.5–2 %, seal 0.7–1.3 $\times$ and flow 0.45–1.6 SLPM without oscillation. $K_P = -12$ began to swing the flow at low flow.
 - **Mode A is unchanged** ($-9.3 / 8.8e-4$). It was already adequate.
- **The model-derived K_P is 10–20 $\times$ the provisional -0.4 .** The provisional gains would regulate very slowly: an upset of 0.1 % O₂ would move the flow by only 0.04 SLPM at first.
- **Noise check for the high gain.** Sensor noise is $\sim 0.001\%$ per sample, and less after the 10-sample mean. $\times 9.3$ that is ≤ 0.01 SLPM, at the MFC's resolution, so the high gain is noise-safe.
- These are **starting points for the bump tests, not final values.** Treat them as an upper bound on aggressiveness until 6.2 is done.
- **Handoff transient.** At handoff the sensor still reads $\sim 0.15\%$ high and settles over ~ 1 min (2.1.5). The loop will see O₂ falling below target just after FBON = 1 and trim the flow down. That is harmless, but it is not a disturbance: consider holding FBON off for ~ 2 min after the handoff flow is reached.

6.0.1 Bump-test duration (DERIVED; changes 6.1)

- With $\tau = 169$ min (mode A) and 53 min (mode B), "wait until O₂ is flat again (30–60 min)" is not achievable: a flat settle needs $\sim 3\tau$, i.e. 8 h and 2.7 h.
- **Instead, identify the process as integrating.** After a flow step ΔF , O₂ bends away from its previous trend after θ (~ 80 – 95 s). Its new slope is $\Delta dC/dt \approx -\Delta F \cdot C/V$ (mode A). Read k' and θ from 20–30 min of data. That is enough for SIMC.
- A full settle is only needed to confirm K. Mode B's K includes the flow-dependence of its ingress, so it is worth one long step.

6.1 Bump test (per mode, feedback off)

1. Settle at the mode's flow until O₂ is flat within 0.01 % over 10 min.
2. Log time, setpoint and O₂ at 1 Hz throughout.
3. Step the flow up by $\sim 15\%$ (at least 0.03 SLPM). Record 20–30 min and fit θ and the new slope, treating the process as integrating (6.0.1). A flat settle takes $\sim 3\tau$: 8 h in mode A, 2.7 h in mode B.
4. Step back down and wait again.
5. Read off gain K, dead time θ and time constant τ . Then:
 - $K_C = \tau / (|K| \cdot (\tau_C + \theta))$
 - $\tau_I = \min(\tau, 4(\tau_C + \theta))$
 - $K_P = -K_C$
 - $K_I = 1/\tau_I$

6.2 First closed-loop trial

1. Step the target down 0.1 %.
2. Expect 63 % of the response in $\sim \tau_C + \theta$, and less than 10 % overshoot.
3. Adjust:

- oscillation with a period of ~40: halve |KP|
- sluggish response: raise |KP| by 1.5×
- slow wander: halve KI

Change one knob at a time.

7. Operator interface

`$(PP)` = 15IDC:SampleGas: in production and SIM:SampleGas: on the bench (section 9.2).

7.0 Screen hierarchy (Phoebus .bob; requested 2026-09-25)

Three levels. Every screen takes the station prefix as a macro, so the same screen files serve 15IDC and 15IDE (7.4).

Level	Audience	Content
Main panel	Users	Large readouts: O2 % , flow (SLPM) , flow setpoint , helium cylinder . The cylinder readout shows litres left from the MFC totalizer, the run-out range (5.5), and pressure once the CYL PV exists. A "New He cylinder fitted" button. State and in-range indicator. Alarm banner showing the current LastAction/alarm text. O2 and flow trend plot. Buttons: Purge , Flow Zero , Resume PID (enabled only in OPEN_LOOP). Selectors: enclosure mode , O2 target . A button opens Admin.
Admin	Beamline staff	Tolerance and Δ. Purge settings: flow, timeout, drop check. PID gains and output limits per mode. Override log and counter. Alarm thresholds. A button opens Deep admin.
Deep admin	Instrument scientist	Every physical and threshold number, as an editable, autosaved field: mode base flows and exponents n; lid threshold, filter time and LidSlope; drop-check time, MinDrop and skip level; ambient O2; ramp-rate clamps (2 / 5 SLPM/s); hard flow ceiling; flow-mismatch tolerance and margin; hold-monitor timing and retries; O2 frozen and invalid limits; averaging count; epid scan and ODEL. Also the station's linked PV names (Alicat prefix, O2 PV, cylinder PV), shown read-only because they are set at IOC start.

- Nothing physical is hard-coded in the SNL program. It reads every number from these records. Each parameter record carries DRVL/DRVH so a typo cannot set, e.g., a 200 SLPM ceiling.
- Phoebus has no per-screen login. "Admin" and "Deep admin" are separate displays reached by button. If stronger protection is wanted, EPICS access security can restrict writes to the deep-admin records by host or user.

7.0.1 Cylinder pressure

- The PV **does not exist yet**. It is passed in as the macro CYL (e.g. CYL=15IDC:HeCy1:Pressure).
- If CYL is empty or not connected, the main panel shows "n/a" and nothing depends on it.
- A future end-state row, "cylinder pressure low", can be added once the PV exists. It would also let the flow-mismatch alarm tell "cylinder empty" apart from "MFC fault".

PV	Type	Purpose
<code>\$(PP)Mode</code>	mbbo	A / B
<code>\$(PP)Target</code>	ao	% O2, default 0.99
<code>\$(PP)Tolerance</code>	ao	% O2, default 0.02. Sets the "in range" indicator only, not a deadband.
<code>\$(PP)PurgeDelta</code>	ao	% O2, default 0.15
<code>\$(PP)Purge</code>	bo	Start purge
<code>\$(PP)FlowZero</code>	bo	Set flow to zero, feedback off

PV	Type	Purpose
\$(PP)ResumePID	bo	OPEN_LOOP → HANDOFF
\$(PP)State	mbbi	Current state
\$(PP>LastAction	stringin	Most recent override or transition reason
\$(PP)OverrideCount	longin	Running count of automatic corrections
\$(PP)O2PID	epid	Loop. Gains and limits are loaded from the mode.

Per-mode parameters (base flow, KP, KI, DRVH, alarm multipliers) are autosaved.

7.5 Graphical simulator for spec review (built 2026-09-25)

simulator/sample_gas_simulator.html is a single file that opens in any browser. It is an executable version of this spec, not IOC code. It contains:

- the §2.1 plant model and the Alicat behaviour (ramp, hold/SDIS, resolution, cylinder)
- the §4 state machine and the exact epid algorithm
- the three screen levels from §7.0 and two independent stations
- world controls, trend plots, an event log and 17 scenario presets from §8
- a time scale of 1× to 3000×

Findings from the first scenario runs, to be decided:

1. **Purge timeout vs Δ (confirms 4.2).** From air, the 6.5 min timeout fires before $O_2 < \text{target} - \Delta$. The "purge incomplete" MINOR alarm is therefore raised on essentially every purge from air.
2. **Alarms need a debounce.** Without one, "O2 above range" toggled MINOR↔MAJOR every second at a threshold. The simulator uses `alarmDelay` = 10 s (the level must persist; this is like the EPICS HYST/delay pattern). Proposed for the real records.
3. **Flow/O2 alarms need a settle time.** (*Superseded by the settling logic in 5.1.1.*) A legitimate target change (0.99 → 0.50 %) saturates the PID at DRVH for tens of minutes. That raised "flow $\geq 2\times$ expected", "PID pinned" and "O2 abnormally high" as MAJOR alarms with nothing wrong. The simulator suppresses these for `alarmSettle` = 30 min after feedback-on or a target change. For a 0.99 → 0.50 % change in mode A that is still too short. Choose the value, or make the flow alarm relative to the PID's own demand.
4. **Open-box drop margin is thin (model assumption).** With the lid open, a 20 SLPM purge drops the reading by ~0.65 % in 60 s in the model, against `MinDrop` = 1.0 %. The open-lid purge test (§8.3) is essential before trusting the drop check.
5. **The wrong enclosure mode looks like a broken PID.** With collimator mode selected and the normal lid fitted, the PID correctly asks for less flow but is held at mode B's DRVH (0.42 SLPM). O2 then drifts down for hours. This is what the user saw in the first simulator, where the fitted lid and the mode were separate selectors. Fixes:
 - the simulator's fitted lid now follows the mode by default
 - a new MINOR alarm fires when the PID is pinned at minimum flow with O2 below target (§5)
6. **Mode-A recovery after a purge is slow.** (Largely fixed by the lag-corrected handoff, 4.2.)
7. **Low-flow regime is now in use but unmeasured.** With `DRVL` = 0.01 SLPM the controller can run between 0 and 0.2 SLPM. There the ingress moves from the with-flow power law to the ~4× higher flow-off leak-in, as over-pressure is lost. The simulator blends the two smoothly, with an unmeasured scale `Fb1end` = 0.05 SLPM. Targets above ~2 % on the normal lid live in this regime, and one bump test there would pin it down. After handoff O2 sits ~0.2 % below target. The PID holds flow at `DRVL` (0.125 SLPM) and O2 takes about an hour to rise back to target. This saves helium, but it is worth knowing.

8. Testing strategy

"Written correctly" means proven against a simulator before it touches the enclosure.

1. Simulation bench, no hardware.

- A caproto soft IOC impersonates the Alicat PVs (setpoint, ramp, hold, `Flow_RBV`) and the O2 analyzer. It uses the leak/dilution model with dead time and noise.
 - The real IOC (`epid` + SNL) runs against it.
 - Scripted scenarios cover every row of the end-state table: normal purge, open lid at start, lid opened in REGULATE, analyzer freeze, MFC hold, bad ramp rates, empty cylinder, IOC restart mid-REGULATE.
 - **Simulator parameters (MEASURED, 2.1):**
 - Volume: $V = 41$ L.
 - Ingress: $J_A = 2.6 \cdot F^{0.04}$ and $J_B = 7.1 \cdot F^{-0.87}$ mL O₂/min. Draw a per-closure seal factor from $\times/\div 1.3$.
 - Flow off: $k = 0.00142$ /min toward ambient (use for both lids until B is measured).
 - Transport delay: 80 s at hold flow, 10 s at 20 SLPM.
 - Sensor-zone lag: first-order, $\tau = 65$ s, applied to the bulk C.
 - Lid lift: exponential to ambient with $\tau = 5.75$ s.
 - Ambient: 19.5 % relaxing to 19.2 % over 1.5 h, plus an occasional -0.6 % dip lasting ~ 1 min.
 - Noise (REVISED 2026-09-25, 2.1.10): white 0.00084 % per 1 Hz sample, plus a slow measurement-side component (Ornstein–Uhlenbeck, relative $\sigma 7.3e-4$, $\tau 79$ s; not bulk O₂). Previously 0.001 % white only.
 - **Extra scenarios from the data:**
 - Operator Flow Zero with the lid kept on for 22 min (must not trip the lid detector).
 - A lift 76 s after Flow Zero, open 2 min, then Purge (24 Sep 11:52).
 - Collimator lid regulated at the normal-lid flow (O₂ climbs slowly; flow-high alarm, not OPEN_STOP).
 - Flow off for 7 h, lid on (O₂ creeps past 10 %; must not trip OPEN_STOP).
 - **Validation target:** the simulator must replay the archived 24 Sep 10 s windows (00:59–04:59, 05:46–13:46) within a few % when fed the recorded flow.
- ### 2. SNL build.
- Compile with `snc` in a real synApps build. Nothing is declared done until it compiles and passes the bench.
- ### 3. Hardware, supervised.
- Bump tests per mode, feedback off.
 - A PID ceiling test for surface vibration.
 - The first closed-loop trial.
 - A deliberate lid-open test.

9. Open decisions

1. Host IOC: DECIDED 2026-09-25.

- **Implementation:** a new synApps soft IOC, `epid` + SNL, separate from the Alicat process.
- **Names:** `procServ/start_ioc` name `15LSS_sample_gas`, port 20125. The compiled app name must start with a letter.
- **Production:** the Linux soft-IOC host, under `$SUPPORT/ChemMat/iocBoot/`, following the existing `start_ioc` conventions.
- **Restart:** `procServ` runs with `--noautorestart`, and `crontab` starts IOCs after a reboot. An IOC crash therefore needs a manual restart. Meanwhile the Alicat holds its last flow and there is no regulation or lid detection. See the end-state row "Controller IOC down".
- **Development:** the MinGW bench on the Windows PC (base 7.0.8.1, seq, std, calc, autosave, asyn).

2. **PV prefix \$(PP): DECIDED 2026-09-25 as 15IDC:SampleGas:** (e.g. 15IDC:SampleGas:State, 15IDC:SampleGas:02PID). This shares the 15IDC: namespace with live beamline PVs, so the bench and simulator **never** use it. They run as SIM:SampleGas: against simulated SIM:Alicat1: and SIM:02 on localhost. The real prefix is applied only at deployment.
3. **Owner sign-off** for writes to Alicat Setpoint, RampRate and Run: **given** by the user, who owns the Alicat IOC. No live writes happen until the simulator-only rule is lifted.
4. **PROVISIONAL values to confirm on hardware** (status after the 2.1 data):
 - MinDrop 1.0 %: a closed box drops 6.8–7.4 %. **Still needs the open-box purge measured.**
 - Ambient O2 19.4 %: **MEASURED** 19.0–19.6 %, relaxing after a lift; a model value of 19.4 % is fine.
 - Lid threshold 10 % for 5 s: **supported** (4.6.1; n=2 lifts, normal lid).
 - LidSlope 0.2 %/s: **supported**, 7× below lifts and 51× above sealed (n=2, normal lid).
 - Flow-mismatch tolerance: open.
 - PID ceiling: open.
 - Gains: model starting values in 6.0, 10–20× the provisional KP. Confirm by bump test.
 - Alicat ramp-rate-0 semantics: open.
5. **Purge timeout vs Δ** (4.2): with target 0.99 % and Δ 0.15 %, reaching target – Δ from air takes 6.4–6.6 min, about the 6.5 min timeout. Raise the timeout (~7.5 min), reduce Δ , or accept the routine alarm.
6. **Setpoint_RBV reliability: RESOLVED 2026-09-25.** The user confirms the live PV is reliable, and the mismatch rule uses it (5.1). Original note (2.1.9): the archiver shows it missing 25 of 41 purges and one flow-off, and reading mid-ramp values. Before relying on the flow-mismatch rule (5.1), confirm with the IOC owner whether the PV itself or only its archiving misses events. If the PV misses them, compare Flow_RBV against the value the sequencer last wrote instead.
7. **Mode-B expected flow:** adopt the exponent $n = 0.53$ (5.2) or keep the linear rule and let the PID absorb the error.
8. **Collimator-lid fast data:** no 10 s record exists for mode B. Capture one lift and one purge at 10 s (or 1 Hz) before finalising mode-B thresholds.
9. **Archiving:** record 15IDC:D1Dmm_calc and Flow_RBV at ≤ 10 s, and the setpoint command (not only the RBV). The 600 s archive missed a whole lid cycle.